

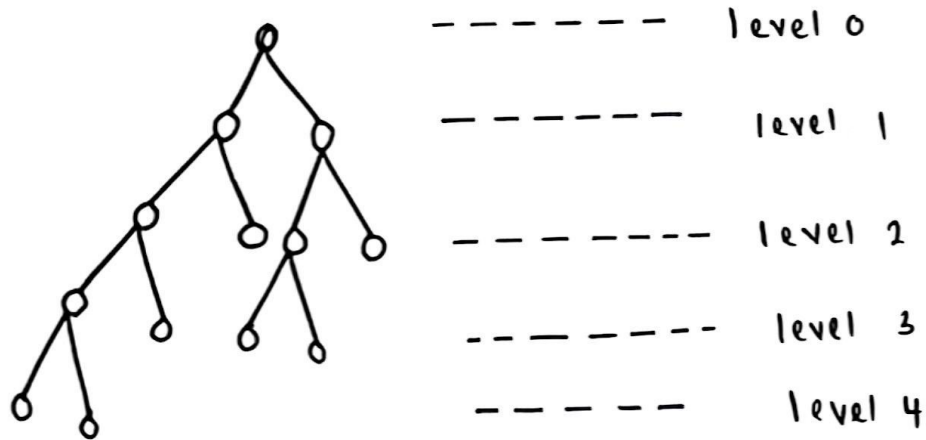
Group members :

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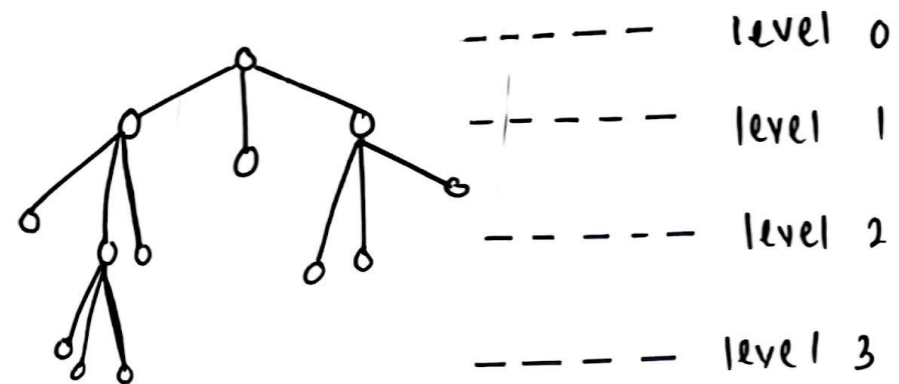
- ①
- a) Not a tree. Because it is a disconnected graph. A tree with m - vertices has $m-1$ edges. The graph given has 6 - vertices and 4 - edges which is not the required 5 edges.
- b) Not a tree. Because it has closed circuit. A tree with m - vertices has $m-1$ edges. The graph given has 6 - vertices and 6 - edges which is not the required 5 edges.
- c) Not a tree. Because it has closed circuit. A tree with m - vertices has $m-1$ edges. The graph given has 6 - vertices and 6 - edges which is not the required 5 edges.

(2) Binary tree of height 4 and full 3-ary tree of height 3

Binary tree of height 4

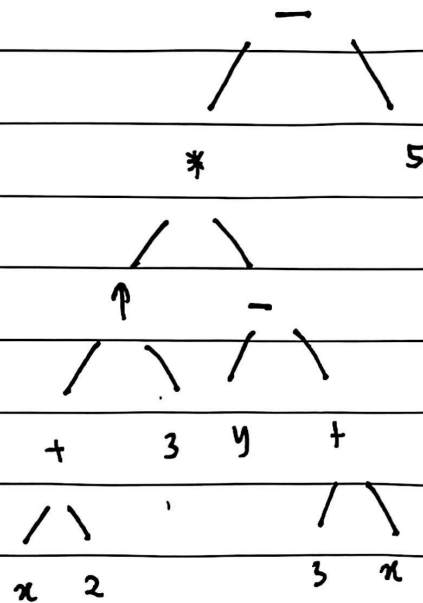


full 3-ary tree of height 3



3. $((x+2) \uparrow 3) * (y - (3+x)) - 5$

Binary tree:



Pre order : -, *, ↑, +, x, 2, 3, -, y, +, 3, x, 5

In order : x, +, 2, ↑, 3, *, y, -, 3, +, x, -, 5

Post order : x, 2, +, 3, ↑, y, 3, x, +, -, *, 5, -

④

$$521 - - 14 ++ *$$

$$\underline{521} - - 14 ++ *$$

$$= \underline{51} - 14 ++ *$$

$$= 4 \underline{14} ++ *$$

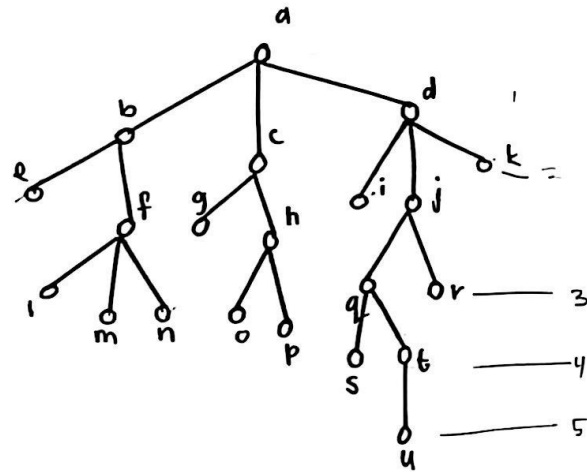
$$= 4 \underline{5} + *$$

$$= 9 *$$

$$= 9 * 9$$

$$= 81$$

5)



(a) vertex is root

= a

(b) vertices are internal node

= a, b, c, d, f, h, j, l

(c) vertices are leaves

= e, g, i, k, m, n, o, p, r, s, u

(d) vertices are children of j

= r, s

(e) vertex is the parent of h

= c

(f) vertices are siblings of o

= m, n

(g) vertices are ancestors of m

= a, b, f

(h) vertices are descendants of b.

= b, f, m, n

6. 5-ary tree with 100 internal vertices

$$i = \frac{n-1}{m}$$

$$100 = \frac{n-1}{5}$$

$$500 = n-1$$

$$n = 501 \text{ vertices} *$$

⑦

$$m = 4 \quad n = 1000$$

$$l = \frac{(m-1)n + 1}{m}$$

$$= \frac{(4-1)1000 + 1}{4}$$

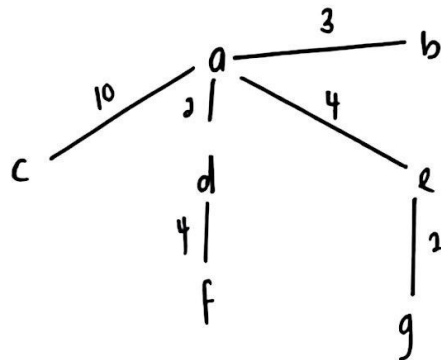
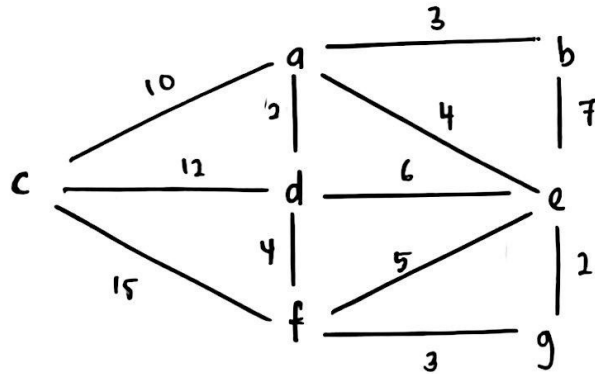
$$= \frac{3001}{4}$$

17

$$= 750.25$$

$$\approx 750 \text{ leaves}$$

8)



List the edges in order in size:

$$ad = 2$$

$$eg = 2$$

$$ab = 3$$

$$fg = 3$$

$$df = 4$$

$$ae = 4$$

$$fe = 5$$

$$de = 6$$

$$be = 7$$

$$ac = 10$$

$$cd = 12$$

$$cf = 15$$

$$\begin{aligned}
 &= ac + ad + eg + ab + df + ae \\
 &= 10 + 2 + 2 + 3 + 4 + 4 \\
 &= 25
 \end{aligned}$$

total weight of tree : 29.

9. - The person in first got \$1 from previous recipient of the letter in that round.

i) Each person sent 6 letters to 6 people : \$ 6

ii) The 6 people each sent another 6 letters to 6 people : \$ 36

iii) Total from (ii) which is \$ 36 is sent to 6 people : \$ 216

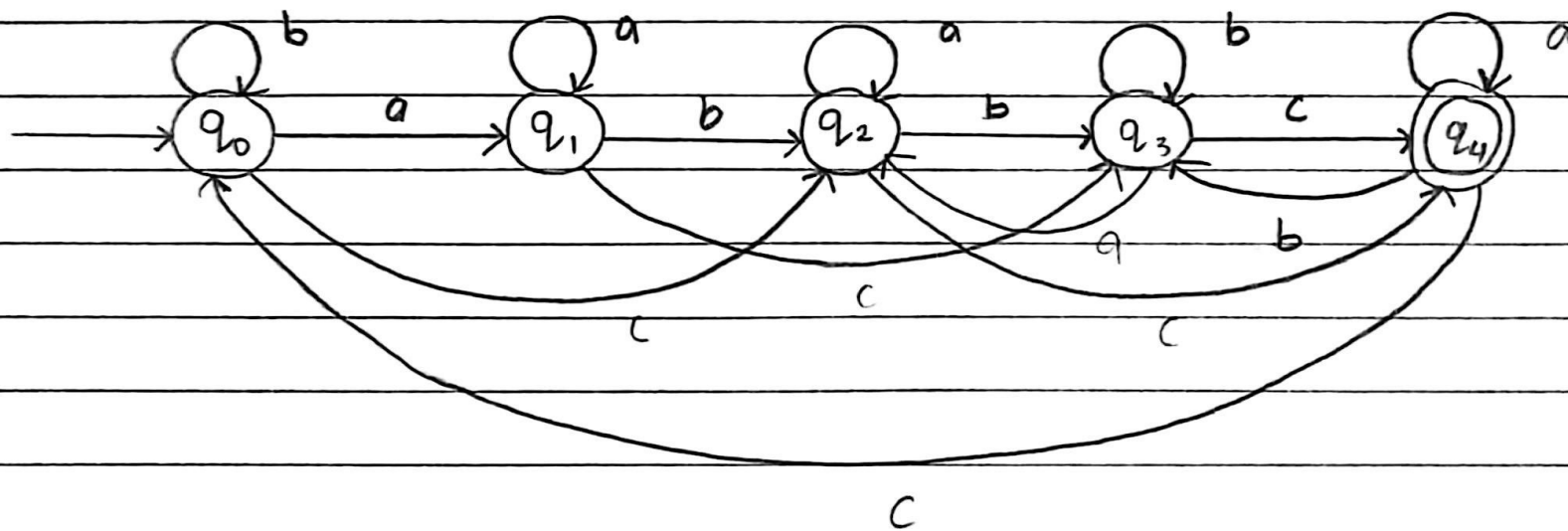
iv) Each will get \$ 216

$\therefore$ At least every person will get \$ 216 .

⑩ accepts $abbc$

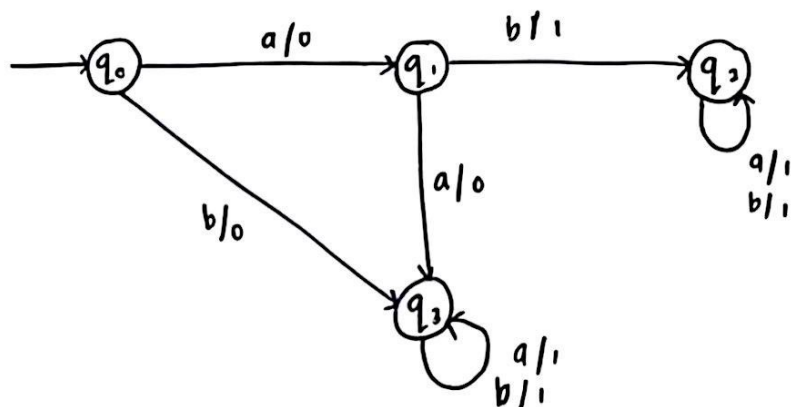
$S = \{q_0, q_1, q_2, q_3, q_4\}$

$I = \{a, b, c\}$



Construct a state diagram of a FSM that accepts the given set of strings over $\{a, b\}$:

(a) contain exactly two 'b's .



Example :-

1) abab

$q_0 \xrightarrow[a/0]{a} q_1 \xrightarrow[b/1]{b} q_2 \xrightarrow[a/1]{a} q_3 \xrightarrow[b/1]{b} q_2$

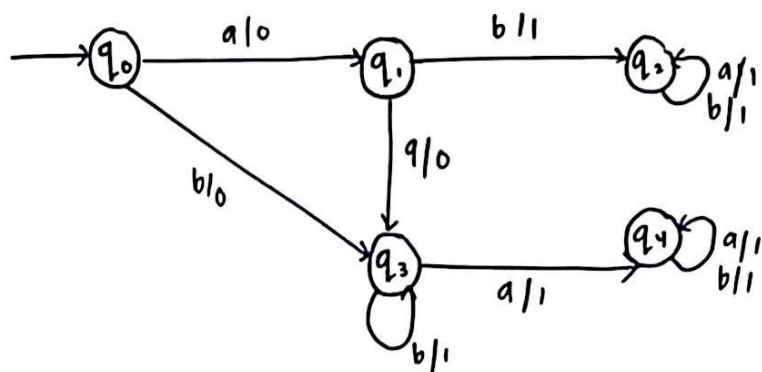
output : 1 , accepted by M.

2) ba ba

$q_0 \xrightarrow[b/0]{b} q_2 \xrightarrow[a/1]{a} q_3 \xrightarrow[b/1]{b} q_2 \xrightarrow[a/1]{a} q_3$

output : 1 , accepted by M.

b) at least one b.



example :-

1) ba'aa

$q_0 \xrightarrow[b/0]{b} q_3 \xrightarrow[a/1]{a} q_4 \xrightarrow[a/1]{a} q_4 \xrightarrow[a/1]{a} q_4$

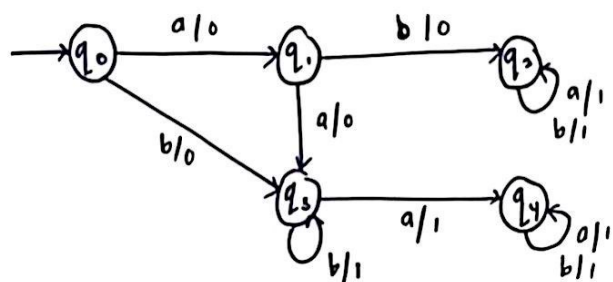
output : 1 , accepted by M.

2) aaaaa

$q_0 \xrightarrow[a/0]{a} q_1 \xrightarrow[a/0]{a} q_3 \xrightarrow[a/1]{a} q_4 \xrightarrow[a/1]{a} q_4 \xrightarrow[a/1]{a} q_4$

output : 1 , accepted by M.

c) odd number of a



example :-

1) bbaaaa

$q_0 \xrightarrow[b/0]{b} q_3 \xrightarrow[b/1]{b} q_3 \xrightarrow[a/1]{a} q_4 \xrightarrow[a/1]{a} q_4 \xrightarrow[a/1]{a} q_4 \xrightarrow[a/1]{a} q_4$

output : 1 , accepted by M.

2) bbbba

$q_0 \xrightarrow[b/0]{b} q_3 \xrightarrow[b/1]{b} q_3 \xrightarrow[b/1]{b} q_3 \xrightarrow[b/1]{b} q_3 \xrightarrow[a/1]{a} q_4$

output : 1 , accepted by M.

12.

(a)

	f_s					f_o				
	i_1	i_2	i_3	i_4	i_5	i_1	i_2	i_3	i_4	i_5
q_0	q_1	-	-	-	-	0	-	-	-	-
q_1	-	q_0	-	-	q_2	-	0	-	-	0
q_2	-	q_0	-	-	q_3	-	0	-	-	1
q_3	-	-	q_4	-	-	-	-	2	-	-
q_4	-	-	-	q_0	-	-	-	-	3	-

(b)

